

Doc. 314 – The Lattice in Hilbert Space: Geometry, Spectrum and Fibre

Johann Pascher

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Doc. 314 – The Lattice in Hilbert Space

What this is about

FFGFT computes geometrically on T^4 with the D4 lattice and quantum-mechanically on $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$ (Docs. 230/282). The transition raises an obvious question: *do the lattices become flat?* Does anything remain of kissing number 24, packing density and triality, or is every lattice the same in Hilbert space?

The answer: the torus is flat anyway – curvature was never the difference – and the lattice structure is not lost but **migrates into the spectrum and into the fibre**. This document computes that through (three scripts, Ch. J) and finds:

1. **The spectrum separates what curvature does not [K]:** Z^4 starts at $|k|^2 = 1$ with degeneracy 8, D4 at $|k|^2 = 2$ with degeneracy 24; odd norms are entirely absent for D4. The first excitation carries exactly $24 = 12 + 12$ – FCC half and winding half separately addressable.
2. **Triality is built in, not attached [K]:** $|\text{Aut}(D4)| = 1152 = |W(F_4)|$, the quotient to $W(D4)$ is $6 = |S_3|$; among the 80 order-3 elements, 16 generate exactly the T^4/Z_3 orbifold with $|\det(1 - A)| = 9$ fixed points and act on every mode orbit as a $\mathbb{C}^3$ circulant.
3. **A negative finding with theorem character [K]:** 5 does not divide $1152 = 2^7 \cdot 3^2$, so $\text{Aut}(D4)$ contains *no* element of order 5. The fivefold rotation from which $\theta = 2/9$ arises (Doc. 293) is in principle incompatible with the lattice – the phase belongs to a different layer.
4. **The doubling is nevertheless prepared [K]:** $24 = 8 \times 3 = 4 \times 6$; four disjoint six-blocks with exactly the C_3 content that $3 \oplus 3'$ requires.
5. **The reciprocity is mode-wise exact – as a place relation [K]:** $\lambda_4 \cdot m = 2\pi$ identically for every mode (de Broglie along the mass direction; winding number = mass number). The *temporal* reading carries $1/\gamma$, a mass mixture a Jensen excess ≥ 1 : the relation becomes exact only for the resting, sharp mode.
6. **The perturbation calculation is carried out [K]:** multiplet sizes follow the irrep dimensions of the respected symmetry, not the orbits – $9 + 8 + 4 + 2 + 1$ in the fully symmetric case, $8 \times 1 + 8 \times 2$ for Z_3 , generically 24 individual levels.
7. **The closure fork is spectrally invisible [K]:** in the rolled-up state there are no fractal corrections – winding numbers are topological, the defect $d = 100\xi$ accumulates only when unrolled. Cases A/B/C of Docs. 313/295 leave the mode spectrum untouched, so Chapters E–H are case-independent. The same boundary settles the fractal question (Ch. J): K_{frak} is cumulative (Doc. 133) and acts only through the scale bridge; locally only $D_f^{\text{space}} = 3 - \xi$ acts, with 6.7×10^{-5} .

Not part of the A-series; status: working document. References: Docs. 230/282 ($\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$), 133 (derivation of K_{frak} : cumulative, RG run), 285 (doubling $6 = 3 \oplus 3'$), 291 (channel analysis), 293 ($\theta = 2/9$ icosahedral), 295/313 (closure fork), 306 (native time-energy reciprocity), 311 (Four onto Three, $24 = 12 + 12$).

Reader's guide: the thread of the argument

The question. In Hilbert space the lattice becomes the index set of the Fourier modes. It is then no longer a sphere packing – in that sense every lattice becomes “flat”. The question is whether the information disappears with it.

The answer in one line. *Geometry* $\rightarrow$ *spectrum* + *fibre*. The degeneracies of the Laplacian are the theta series of the lattice; the kissing number becomes the degeneracy of the first excitation; triality becomes the orbifold and fibre structure.

What follows. First, two flat tori are distinguishable by their spectra (Ch. B). Second, the $\tilde{T} \cdot m = 1$ structure becomes spectrally readable when $R_4 \neq R_{123}$, and the level ordering changes at exactly two root radii (Ch. C). Third, the Z_3 need not be imposed – it is a lattice symmetry (Ch. E).

The boundary, drawn cleanly. Fivefold symmetry is impossible on the lattice (Ch. F). This does not hit the doubling, because that lives on the *fibre* and the interface of the two levels is exactly C_3 – occupied by triality (Ch. G). And the container structure holds on *every* shell, so it cannot distinguish one (Ch. H) – the same pattern-versus-occupation logic as in Doc. 293.

The caveat. The computation uses the smooth Laplacian, i.e. $D_f = 4$ exactly. All numbers are therefore *bare*; Ch. I separates what that costs: counts and ratios stay invariant, $D_f^{\text{space}} = 3 - \xi$ is *local* and acts with 6.7×10^{-5} , while $K_{\text{frak}} = 1 - 100\xi$ is *cumulative* (RG run across 17 orders of magnitude) and acts only through the scale bridge. Bare are therefore only the SI absolute values (–1.33 % via the anchor); counts, ratios and containers are not. The degeneracy splitting under perturbation is computed – it follows the irrep dimensions of the perturbation symmetry, not the orbits, and lies at $\sim 10^{-5}$. Ch. D2 adds the boundary between rolled-up and unrolled states, which is why the closure fork (Docs. 313/295) stays spectrally invisible.

In one sentence: in Hilbert space the lattice becomes flat but not inconsequential – what geometry carried is carried by spectrum and fibre, and the division of labour between them is exactly determinable, as long as the smooth approximation is booked as such.

Notation

The table explains the symbols of this document. Quantities appearing only once are introduced there.

Note on notation. In lattice theory Λ is the usual symbol for a lattice (Conway/Sloane). In the FFGFT corpus, however, Λ is taken – as the cosmological constant, in Λ CDM, and in particular as the closure scale $\Lambda^* = 1/R_H^2$ (Doc. 312). This document therefore uses Γ throughout for the lattice and Γ^* for the dual lattice. Likewise L^* is *not* used here, since in Doc. 312 it denotes the cycle length; the torus edge lengths are called R_{123} and R_4 .

Symbol	Meaning
<i>Lattice and torus</i>	
Γ	lattice in $\mathbb{R}^4$; defines the torus $T^4 = \mathbb{R}^4/\Gamma$ (instead of the Λ customary in the lattice literature, see note on notation)
Γ^*	dual lattice; carries the admissible wave vectors of the Fourier modes
D_4	root lattice D_4 : all integer k with $\sum k_i$ even; kissing number 24

Symbol	Meaning
Z^4	cubic lattice Z^4 ; kissing number 8, exactly self-dual
$D4^*$	dual D4 lattice; again a D4 lattice up to scaling
R_{123}, R_4	radii of the three spatial directions and of the fourth direction
r	radius ratio $r = R_4/R_{123}$ (dimensionless)
<i>Spectrum</i>	
k	wave vector (mode index), $k \in \Gamma^*$
k_4	fourth component of k ; $k_4 = 0$ separates the FCC half from the winding half
$ k ^2$	norm; also the eigenvalue of the Laplacian on the isotropic torus
$E(k)$	eigenvalue under anisotropy, $E = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$
$N(n)$	degeneracy of the shell $ k ^2 = n$; coefficient of the theta series
θ_{D4}	theta series of the D4 lattice, $\theta_{D4} = (\theta_3^4 + \theta_4^4)/2$
$\theta(t)$	heat-kernel theta function, $\theta(t) = \sum_k e^{-t k ^2}$
$\zeta_M(s)$	Epstein zeta function of the lattice M ; $s = -1/2$ is the Casimir point
<i>Symmetries</i>	
$\text{Aut}(\Gamma)$	automorphism group of the lattice (length-preserving self-maps)
$W(F_4), W(D4)$	Weyl groups; $ W(F_4) = 1152$, $ W(D4) = 192$
A	element of order 3 in $\text{Aut}(D4)$; half-integral, hence computed exactly via $2A$
$\det(1 - A)$	determinant; $ \det(1 - A) = 9$ counts the orbifold fixed points
Z_3, C_3	cyclic group of order 3 (triality resp. fibre action)
S_3, A_5	symmetric group on 3 elements; alternating group on 5 elements (icosahedral)
ω	third root of unity, $\omega = e^{2\pi i/3}$; phase $+120^\circ$
χ_0, χ_1, χ_2	the three Z_3 characters; index the orbifold sectors
<i>Hilbert space and fibre</i>	
$\mathcal{H}$	state space, $\mathcal{H} = L^2(\Gamma^4) \otimes \mathbb{C}^3$ (Docs. 230/282)
$\mathbb{C}^3$	fibre over each point; carries the Z_3 structure
$3, 3'$	the two triplets of the doubling $6 = 3 \oplus 3'$ (Doc. 285)
φ	golden ratio, $\varphi = (1 + \sqrt{5})/2$; occurs only in the A_5 layer
$\theta = 2/9$	phase share from Doc. 293; not a lattice invariant (Ch. F)
<i>FFGFT references</i>	
$\xi = 4/30000$	geometric fundamental constant of FFGFT
$\tilde{T} \cdot m = 1$	time-mass relation; readable in the spectrum as the splitting at $r \neq 1$
<i>Status labels</i>	
[K]	core derivation: computed and controlled

Symbol	Meaning
[S]	structural statement: consistent but not derived
theorem	mathematically proven, not merely checked numerically
boundary	explicitly booked limitation of scope

Chapter A

The starting position: flatness is not the difference

The torus $T^4 = \mathbb{R}^4/\Gamma$ is *always* flat, whatever lattice Γ defines it. The D4 torus and the Z^4 torus both have zero curvature. The difference lies not in curvature but in lattice structure: kissing number 24 against 8, packing density $\pi^2/16$ against $\pi^2/32$.

On passing to $\mathcal{H} = L^2(T^4)$ the lattice becomes the **index set of the Fourier modes**: the basis consists of plane waves $e^{ik \cdot x}$, and the admissible k lie on the dual lattice Γ^* . In this sense every lattice becomes flat in Hilbert space – it is no longer a sphere packing in space but a point lattice of quantum numbers.

The geometry is not lost; it migrates. The Laplacian has eigenvalues $|k|^2$, and the degeneracy of each eigenvalue is the number of lattice vectors of that length – the theta series of the lattice. The 24 shortest D4 vectors become the 24-fold degeneracy of the first excitation, and the decomposition $24 = 12+12$ of Doc. 311 (FCC neighbours plus winding neighbours) remains exactly readable as a k_4 split.

Two anchors keep this translation stable:

1. **D4 is self-dual up to scaling** – $D4^*$ is again a D4 lattice. The Fourier transform does not lead out of the family. (Z^4 is exactly self-dual.)
2. **The fibre changes nothing about the lattice.** In $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$ the $\mathbb{C}^3$ fibre hangs at every point and carries the Z_3 structure; the mode lattice remains Γ^* .

$$\boxed{\text{geometry} \longrightarrow \text{spectrum} + \text{fibre}}$$

(A.1)

Chapter B

The spectrum separates what curvature does not [K]

Complete shell count up to $|k|^2 = 12$:

Norm $ k ^2$	Z^4	D4	D4 split ($k_4=0$ / $k_4 \neq 0$)
1	8	0	—
2	24	24	12 / 12
3	32	0	—
4	24	24	6 / 18
5	48	0	—
6	96	96	24 / 72
7	64	0	—
8	24	24	12 / 12
9	104	0	—
10	144	144	24 / 120
11	96	0	—
12	96	96	8 / 88

Three readings:

(i) Two flat tori are spectrally distinguishable. Z^4 starts at norm 1 with degeneracy 8, D4 at norm 2 with degeneracy 24. For D4 the odd norms are *entirely* absent – the parity condition $\sum k_i$ even excludes them. What curvature does not distinguish, the spectrum distinguishes at once.

(ii) Independent control passed. The D4 series 24, 24, 96, 24, 144, 96 ... is the known theta series $\theta_{D_4} = (\theta_3^4 + \theta_4^4)/2$; the script reproduces it coefficient by coefficient.

(iii) The 12 + 12 decomposition is spectrally addressable. The first excitation carries exactly twelve modes with $k_4 = 0$ (the spatial FCC half) and twelve with $k_4 \neq 0$ (the winding half). On higher shells the split is asymmetric (6/18 at norm 4, 8/88 at norm 12): **the 12/12 symmetry is a property of the first shell, not a general theorem.** This must be booked explicitly – Doc. 311 argues with $24 = 12 + 12$, and that statement holds exactly where it is made.

Chapter C

Anisotropy: where the time-mass relation becomes readable [K]

If the fourth direction is compactified with a different radius ($r = R_4/R_{123}$, eigenvalue $E(k) = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$), the first shell splits:

r	E (winding, 12 modes)	E (FCC, 12 modes)
1.0	2.000	2.000 (degenerate: 24)
1.1	1.826	2.000
1.5	1.444	2.000
2.0	1.250	2.000

The FCC modes stay fixed at $E = 2$, the winding modes follow exactly

$$E_{\text{winding}} = 1 + \frac{1}{r^2}, \quad \Delta = 1 - \frac{1}{r^2}. \quad (\text{C.1})$$

The splitting is a direct measure of the radius ratio – **the place where the $\tilde{T} \cdot m = 1$ structure becomes readable in the spectrum**: mass direction and spatial directions carry different excitation energies as soon as they have different lengths.

C.1 The crossing map is discrete and short

All levels have the form $E = a + b/r^2$ with integer a, b . In the range $1 < r \leq 2.5$, among all levels up to norm 8 there are **exactly two** crossing radii:

$r = \sqrt{2}$	pure winding (0, 0, 0, ± 2) crosses the FCC-12
$r = \sqrt{3}$	pure winding crosses the winding-12

The level sequence therefore does not change continuously and chaotically but at two distinguished root radii. Any ordering statement about the lowest excitations falls into exactly three regimes: $r < \sqrt{2}$, $\sqrt{2} < r < \sqrt{3}$, $r > \sqrt{3}$. Beyond $\sqrt{3}$ the *ground tone* of the spectrum is a pure winding (at $r = 2$: $E = 1.00$ with 2 modes before 1.25 with 12 before 2.00 with 12).

Chapter D

Time from the mass circle: the reciprocity is mode-wise exact [K]

The fourth direction is the time–mass dual one; unrolling is onto time. Two check questions: how do time periods arise – and does $\tilde{T} \cdot m = 1$ hold only on average, when mass and time “distribute equally”?

D.1 Time periods as covering-space bookkeeping

Unrolling the compact fourth direction is the universal covering $\mathbb{R} \rightarrow S^1$; the deck transformation group is $\mathbb{Z}$, and the unrolled coordinate counts cycle traversals. A k_4 mode $e^{ik_4 x_4/R_4}$ thereby becomes a periodic function with period $T_k = 2\pi R_4/k_4$. Since $\pi_1(S^1) = \mathbb{Z}$ is abelian, this bookkeeping is *lossless* – the same construction that raises the faithfulness question for non-abelian fundamental groups is faithful here by structure.

D.2 Mode-wise exactness – and what exactly is exact

In the deformation spectrum (Ch. C), $E(k) = k_1^2 + k_2^2 + k_3^2 + (k_4/r)^2$ is literally $E^2 = p_{\text{space}}^2 + m^2$ with $m_k = k_4/R_4$ – **the winding number is the mass number**. Hence

$$T_k \cdot m_k = \frac{2\pi R_4}{k_4} \cdot \frac{k_4}{R_4} = 2\pi \quad \text{exactly, for every mode.} \quad (\text{D.1})$$

The reciprocity therefore does *not* hold only on average – it is satisfied mode-wise identically, because period and mass are built from the same quantum number k_4 and the same radius. The next section makes precise, however, *which* quantity is exact here: T_k is first of all an arc length on the circle, i.e. a place interval. Side result: the 12 + 12 splitting (Ch. C) is, in this reading, the separation massless ($k_4 = 0$, FCC) versus massive (winding), and the level crossing at $r > \sqrt{3}$ means: under strong anisotropy a pure mass mode becomes the ground tone.

D.3 The period is a place first, not a duration

The quantity $T_k = 2\pi R_4/k_4$ is an *arc length on the circle*: the distance in the coordinate x_4 after which the mode repeats. It is therefore a **place interval**, not elapsed time. Properly it is called λ_4 , and the relation

$$\lambda_4 \cdot m = \frac{2\pi R_4}{k_4} \cdot \frac{k_4}{R_4} = 2\pi \quad (\text{D.2})$$

is the **de Broglie relation** along the mass direction – wavelength times wavenumber – not yet a statement about time. It is exact and always valid, but it is a statement about x_4 .

Where, then, does duration come from? Only from the unrolled coordinate. A point of unrolled time is a pair

$$t = w \cdot 2\pi R_4 + x_4, \quad w \in \mathbb{Z} \text{ (winding)}, \quad x_4 \in [0, 2\pi R_4), \quad (\text{D.3})$$

that is, *number of revolutions plus place on the cycle*. Duration resides entirely in the counter w – the deck transformation of the covering – and place in the remainder. On the cycle itself there is no distinguished beginning; every “moment” is a place that recurs (cf. Doc. 313).

D.4 The temporal reading carries a γ factor

If one asks for the *oscillation period in unrolled time*, the relevant quantity is $T_{\text{osc}} = 2\pi/E$ with $E = \sqrt{p_{\text{space}}^2 + m^2}$. Then

$$T_{\text{osc}} \cdot m = 2\pi \frac{m}{E} = \frac{2\pi}{\gamma} \leq 2\pi, \quad \text{equality} \iff p_{\text{space}} = 0. \quad (\text{D.4})$$

p_{space}/m	γ	$T_{\text{osc}}m/2\pi$	$\lambda_4 m/2\pi$
0	1.00000	1.0000000	1.0000000
0.5	1.11803	0.8944272	1.0000000
1	1.41421	0.7071068	1.0000000
2	2.23607	0.4472136	1.0000000
5	5.09902	0.1961161	1.0000000

Place and duration are not the same here. Exactly 2π is the *place* reading; the *time* reading carries the factor $1/\gamma$ and coincides with it only for the mode at rest. Carrying T_k as a “time period” and booking $T_k \cdot m = 2\pi$ as temporal reciprocity would accordingly confuse the two.

Three statements in three directions:

Reading	Relation	Equality for
place (λ_4 , de Broglie)	$= 2\pi$	always
time, moving mode	$= 2\pi/\gamma \leq 2\pi$	$p_{\text{space}} = 0$
mass mixture (Jensen)	$\geq 2\pi$	sharp shell

The exact relation $\tilde{T} \cdot m = \text{const}$ therefore holds for the **resting, sharp** mode – there and only there both correction factors vanish simultaneously. Motion pushes the product down, mass spread lifts it up. This is no contradiction to Doc. 306 but the Kaluza–Klein reading of its validity conditions.

D.5 For mixtures the equation becomes an inequality

If a state superposes several k_4 shells with weights p_k , then $\langle T \rangle \langle m \rangle = 2\pi \langle k \rangle \langle 1/k \rangle$, and by Jensen

$$\langle k \rangle \langle 1/k \rangle \geq 1, \quad \text{equality} \iff \text{exactly one shell occupied.} \quad (\text{D.5})$$

Numerically (script 4): sharp shell (any k) $\rightarrow$ exactly 1.0000000000; $k = 1, 2$ equally occupied $\rightarrow$ 1.125; uniform $k = 1 \dots 50 \rightarrow$ 2.295; thermal $p_k \propto e^{-k/T} \rightarrow$ 1.0000227 ($T = 0.1$) up to 2.600 ($T = 10$); 2000 random distributions: minimum 1.077, never below 1.

Booking – the conjecture inverts: the condition for $\tilde{T} \cdot m = 1$ is *not* equal distribution but **sharpness:** exactly one occupied mass shell. Equal distribution is, on the contrary, a state with a pronounced reciprocity excess (2.29 for $k \leq 50$). The excess $\langle k \rangle \langle 1/k \rangle - 1$ measures the mass spread of the state (for small spread $\approx \text{Var}(k)/\langle k \rangle^2$): a mixture has neither a sharp period nor a sharp mass, and the product of the means lies strictly above 2π .

Together with the γ factor the full condition follows: $\tilde{T} \cdot m$ is constant exactly when the mode *rests* (no γ) *and* is sharp (no Jensen excess). Both deviations have different signs and different causes – motion in space versus spread in mass.

Epistemic status: lattice-side consistency check of the mode structure (Kaluza–Klein reading of the fourth direction); the native FFGFT formulation of the reciprocity is Doc. 306 – nothing is newly derived here; compatibility is checked: passed, with the sharpening “exact per mode, $\geq$ for mixtures”.

D.6 The closure fork is spectrally invisible

Doc. 313 (after Doc. 295) poses the closure question as a fork: the defect $d = 100\xi = 1/75$ per revolution admits three cases for the *unrolled* time. With the cycle length $\tau_c = 2\pi/H_0 = 91.9$ Gyr (Doc. 312):

Case	Condition	Closure	Period
A	$d = 0$	after 1 revolution	91.9 Gyr
B	ξ frozen	after 75 revolutions	6892 Gyr
C	ξ running (recursion)	never	log spiral

Case B follows from the rational rotation number $1 - d = 74/75$: since $\text{gcd}(74, 75) = 1$, the trajectory closes after exactly 75 revolutions, i.e. $75\tau_c = 6892.5$ Gyr. Case C arises when ξ runs through the recursion; the accumulated defect becomes logarithmic and the winding turns into an equiangular spiral.

This table is only quoted here. It stands so that the cases are intelligible without looking them up; its *interpretation* – entropy obligation, Poincaré discrepancy, thermal history – belongs to Docs. 312/313 and is not pursued here. Two points are nevertheless needed so the numbers are not misread.

First: these are deadlines, not repetitions. 91.9 and 6892 Gyr do not say that something repeats, but within what period a recurrence would *have to* occur if the respective case held. What is closed is the *geometry*; whether the *history* returns is independent of that and, per Doc. 313, coarse-grained open. In this reading the “Big Bang” is in any case a *place* on the cycle – the region of smallest masses and slowest clocks – not an event; a closed circle has no starting point.

Second, and this bounds the reach of all time statements: states at other places on the cycle are not derivable from today’s. The usual backward extrapolation ($T(z) = T_0(1+z)$ from an adiabatic chain) presupposes *one* continuous thermodynamic history. In the place reading, an early epoch is not an earlier state of the same system but a location with a *different local structure* – different masses, different clock rates – and hence its own equilibrium, fixed locally and not by continuation of today’s. The corpus demonstrates this on itself: T_0 is not extrapolated but obtained structurally from ξ ($k_B T_0 = (16/9)\xi$, Docs. 061/313). This is precisely why the coarse-grained chaining remains open (Poincaré discrepancy 10^{104} dex, D(ii) in Doc. 313).

What this does *not* mean: it devalues no structural derivation. The Ω_m^* chain in Doc. 313 runs through ξ and K_{frak} , not through an adiabatic back-calculation, and is unaffected by this boundary. Affected are only statements of the form “how hot was it in epoch X” insofar as they are obtained by extending today’s equilibrium – such statements would need their own structural derivation *at that place*.

It would be tempting to read the defect as a *twist* of the boundary condition ($k_4 \rightarrow k_4 + 1/75$, Scherk–Schwarz-like). That is **wrong**, and the reason is structural:

In the rolled-up state there are no fractal corrections. Single-valuedness of the mode on the circle forces $k_4 \in \mathbb{Z}$ – integer or nothing. A winding number is *topological*; it has no room for $1/75$. Corrections require something that *accumulates*, and only a traversed path can accumulate. The very phrase “defect per revolution” presupposes counting revolutions – and revolutions are counted only in the unrolled coordinate w (Ch. D1). The fractal path correction K_{frak} is a property of the *traversal*, not of the periodicity of the field.

Hence the assignment:

Quantity	State	fractal correction
theta series, degeneracies, $24 = 12 + 12$	rolled up	no
$\text{Aut}(D_4)$, triality, orbifold	rolled up	no
circulant, containers 4×6	rolled up	no
Casimir, heat kernel, congruences	rolled up	no
$\lambda_4 \cdot m = 2\pi$ (de Broglie)	rolled up	no
cycle length, defect d , case A/B/C	unrolled	yes
revolution count w , history	unrolled	yes
T_{osc} , γ factor	unrolled	yes

Two consequences. First: *the fork has no spectral signature*. The mode spectrum is an object of the rolled-up state; whether the trajectory closes after one, after 75 or after no revolution changes nothing about the eigenvalues of the Laplacian on T^4 . The first shell carries 24 modes in all three cases.

Second, and this is the real relief: **Chapters E–H are case-independent**. Orbifold, circulant and containers do not depend on which of the three cases holds – they are statements about the compact geometry, and the defect does not act there. The fork is a question of *history* (entropy, recurrence – Doc. 313), not of *structure*.

The same boundary settles the fractal question. D_f and K_{frak} are *different* things (Doc. 133): $D_f^{\text{space}} = 3 - \xi$ is local and practically without effect (6.7×10^{-5}), $K_{\text{frak}} = 1 - 100\xi$ is cumulative across the RG run and acts only where scales are *traversed*. The rolled-up spectrum carries neither appreciably – Ch. J elaborates.

Marginal note on the number. $75 = 1/(100\xi)$; its factorisation $3 \cdot 5^2$ carries *no* meaning – the five here has nothing to do with the fivefold symmetry of Ch. F.

Chapter E

Reorganisations: three levels and one boundary

Flat lattices can be reorganised – on three levels that cost very different things.

Change of basis (free). Any unimodular transformation from $GL(4, \mathbb{Z})$ describes the same point lattice with different generators. The spectrum does not change at all. Pure bookkeeping.

Lattice symmetries (free, structure-bearing). Reorganisations mapping the lattice onto itself – this is where the central finding sits, Ch. E.

Rebuilding into another lattice (costs physics). Discretely via glue vectors along the chain

$$D4 \subset \mathbb{Z}^4 \subset D4^* \quad (\text{index 2 each}), \quad (\text{E.1})$$

or continuously by deformation in the moduli space $GL(4, \mathbb{R})/(O(4) \times GL(4, \mathbb{Z}))$. Each step shifts degeneracies; $D4$ is a distinguished point there (kissing number 24, provably unbeatable in 4D).

The boundary of the spectral statement. From dimension 4 on there exist pairs of flat tori that are *isospectral but not isometric* (Schiemann). The general theorem “the spectrum determines the lattice” is *false* in 4D. Uncritical for $D4$ itself – kissing number 24 characterises it uniquely – but it must not be booked as a rule.

Chapter F

Triality is built in, not attached [K]

F.1 Computed on the lattice

$ \text{Aut}(D_4) $	$1152 = W(F_4) $
$ \text{Aut}(Z^4) $	$384 = 2^4 \cdot 4!$
$ \text{Aut}(D_4) / W(D_4) $	$1152/192 = 6 = S_3 $

Both orders were determined by brute-force enumeration of all basis tuples with identical Gram matrix and match the known values. The quotient $6 = |S_3|$ establishes triality *directly on the lattice automorphism group*, not merely on the Dynkin diagram: D_4 is the only lattice of the D_n family with three equivalent Dynkin arms; for $D_5, D_6, \dots$ the outer symmetry shrinks to Z_2 .

F.2 The 80 order-3 elements in three classes

Computed exactly (the triality elements are *half-integral* – the glue-vector mixture $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ enters; rounding to integers destroys them, hence the computation uses $2A$):

Class	Count	Character
trace -2 , half-integral	16	fixed-point free in $\mathbb{R}^4$; eigenvalues $\{\omega, \bar{\omega}\}$ twice
trace 1, integral	32	fixed plane in $\mathbb{R}^4$, free on the 24 minimal vectors
trace 1, half-integral	32	6 fixed vectors + 6 triple orbits

F.3 Two consequences

(a) The orbifold is a lattice symmetry, not an extra assumption. For each of the 16 elements with trace -2 one has exactly

$$|\det(1 - A)| = 9, \quad (\text{F.1})$$

so the quotient $T^4/\langle A \rangle$ is a Z_3 orbifold with *exactly 9 fixed points* – the standard number 3^2 . All nine lie at third-coordinates. **The T^4/Z_3 structure need not be imposed; it is already present in D_4 .**

(b) On every mode orbit the Z_3 acts as a circulant. The unitary action on $L^2(T^4)$ permutes the three plane waves of a triple orbit cyclically; the permutation matrix has eigenvalues $\{1, \omega, \omega^2\}$, phases $0^\circ, +120^\circ, -120^\circ$ – *structurally exactly the circulant diagonalisation of the $\mathbb{C}^3$ fibre* (Doc. 291). The first shell provides eight copies: $24 = 8 \times 3 = 8 \times (\chi_0 \oplus \chi_1 \oplus \chi_2)$.

Chapter G

The negative finding: 5 does not divide 1152 [K]

The order distribution in $\text{Aut}(D_4)$, determined exactly:

Order	1	2	3	4	6	8	12
Count	1	139	80	228	464	144	96

Total 1152. **Order 5: zero elements** – and that is not a computational result but a theorem: $1152 = 2^7 \cdot 3^2$, and 5 does not divide it (Lagrange).

Consequence for $\theta = 2/9$. The fivefold rotation R_5 , from which $p_0 = |\langle v_0 | R_5 v_e \rangle|^2 = 2/9$ follows (Doc. 293), is *in principle* incompatible with the D_4 lattice structure. The test was additionally carried out directly: all invariants of the linear lattice action at the nine orbifold fixed points ($x \cdot x$, $k \cdot x$, $x \cdot y$, all mod 1 in exact rational arithmetic) have denominator spectrum $\{1, 2, 3, 6\}$ – **no ninths**. Notably, $x \cdot y$ is a multiple of one ninth, but the numerators are always divisible by 3; the D_4 geometry *suppresses* ninths at the first level.

What this means – and what it does not. $2/9$ is not an invariant of the lattice action. This does not devalue Doc. 293 but delimits it: the phase belongs to a layer *above* the lattice (the A_5 structure), the lattice carries the Z_3 . The division of labour is thereby exactly named rather than surmised.

Chapter H

The doubling is nevertheless prepared [K]

Does the negative finding contradict the doubling $6 = 3 \oplus 3'$ of Doc. 285? No – in three specifiable stages.

(1) The prohibition hits the wrong level. $5 \nmid 1152$ forbids order-5 *lattice automorphisms*. The doubling lives on the **fibre** $(\mathbb{C}^3 \rightarrow \mathbb{C}^6)$; A_5 is nowhere required as a spatial symmetry. The compatibility condition is only that the interface of the two levels exists – and that is exactly determinable: element orders in A_5 are $\{1, 2, 3, 5\}$, in $\text{Aut}(D_4)$ they are $\{1, 2, 3, 4, 6, 8, 12\}$; the intersection is $\{1, 2, 3\}$, canonically C_3 . **And exactly that is provided by triality.** The interface is not merely permitted, it is the only possible one – and it is occupied.

(2) The lattice prepares the doubling. The central involution $-1 \in \text{Aut}(D_4)$ commutes with the triality Z_3 and pairs the eight triple orbits of the first shell into **four disjoint pairs**. No orbit is self-paired, provably: $-k = A^j k$ would require eigenvalue -1 for an element of order 3 – impossible. Hence

$$24 = 8 \times 3 = 4 \times 6 \quad (\text{orbit} \oplus \text{mirror orbit}). \quad (\text{H.1})$$

Each six-block carries under C_3 the character content $2 \times (\chi_0 \oplus \chi_1 \oplus \chi_2)$ – exactly the restriction of $3 \oplus 3'$. The first excitation shell provides four ready containers.

(3) The boundary, cleanly booked. The *labelling* – which factor is 3 and which $3'$ – is lattice-blind. The A_5 characters of the two triplets differ only on the 5-classes (φ against $1 - \varphi$); on the 3-class both are zero and the C_3 restrictions are identical. The distinction hangs exactly on the structure that does not exist in the lattice – fully consistent with Ch. F.

Balance: the lattice provides containers and C_3 content, the A_5 layer provides the $3/3'$ labels, φ and the fivefold – and hence $2/9$.

Chapter I

Anchor-free results [K]

All quantities below are dimensionless – ratios, counts, congruences. No Kammerton, no SI scale enters.

I.1 Two congruence theorems for the D4 theta series

All complete shells up to norm 120 checked:

$$N(n) \equiv 0 \pmod{3} \quad \text{and} \quad N(n) \equiv 0 \pmod{6} \quad \text{without exception.} \quad (\text{I.1})$$

Both are theorems, not empirics: the triality Z_3 is fixed-point free on $\Gamma \setminus \{0\}$ (eigenvalues $\omega, \bar{\omega}$ – no eigenvalue 1), so *every* shell decomposes into triple orbits; together with the -1 pairing without self-pairing, into six-blocks.

I.2 Character-resolved T^4/Z_3 spectrum

Consequence: exact equidistribution on every shell.

Norm	$N(D4)$	χ_0	χ_1	χ_2
2	24	8	8	8
4	24	8	8	8
6	96	32	32	32
10	144	48	48	48
14	192	64	64	64
18	312	104	104	104

The invariant sector of the orbifold (χ_0) has exactly $N(n)/3$ states per shell; the Z_3 projection costs no asymmetry on any shell.

Honest consequence. Equidistribution and six-containers are *not* a distinction of the first shell but a theorem for all. **The structure is everywhere – and therefore cannot select anything.** This is exactly the pattern-versus-occupation logic of Doc. 293: the pattern is present scale-invariantly; which shell is physically occupied is decided by something else.

I.3 Casimir: the densest lattice loses

ζ -regularised vacuum energy of a scalar field, $E = \pi \zeta_M(-1/2)$, both tori normalised to co-volume 1 (Epstein zeta via the Riemann split representation, verified three times – against

direct summation at $s = 4, 5$ and against the closed form $Z_{Z^4}(s) = 8(1 - 4^{1-s})\zeta(s)\zeta(s-1)$, agreement to 25 digits):

	$\zeta_M(-1/2)$	$E = \pi\zeta_M(-1/2)$
Z^4 torus	-0.2966893	-0.9320768
D4 torus	-0.2764778	-0.8685805
Difference		+0.0634963

The Z^4 torus lies lower. The reason is the spectral gap: at equal volume the densest lattice D4 has the *largest* first eigenfrequency ($\sqrt{2}$ against 1) and hence the *less* negative zero-point energy. Notable is the reversal of ordering: for $s > 0$ D4 minimises the Epstein zeta (the known extremality of the densest lattice), at $s = 0$ all lattices agree ($\zeta(0) = -1$ universally), and at $s = -1/2$ the ordering is inverted.

Interpretation boundary, booked: this holds for the scalar, periodic field. Different field content or different boundary conditions can change the sign of the preference – “nature chooses Z^4 ” does *not* follow.

I.4 Heat kernel: the observability limit quantified

For both tori $t^2\theta(t) \rightarrow 1$ (identical Weyl term at equal volume); the lattice difference is exponentially small:

t	relative difference
0.5	1.2×10^{-2}
0.2	1.2×10^{-6}
0.1	1.8×10^{-13}
0.05	4.1×10^{-27}

Any observable resting only on the leading heat-kernel coefficients is lattice-blind. The D4/ Z^4 difference sits exclusively in the exponentially suppressed theta-series fine structure – or directly in the spectral gap.

Chapter J

What is still bare here – and what is not [K]

This document computes with the *smooth* Laplacian on $T^4 = \mathbb{R}^4/\Gamma$. What has to be settled is what FFGFT holds against this in fractal structure – and here **two quantities must be kept apart** that are easily confused:

Quantity	Character	Effect
$D_f^{\text{space}} = 3 - \xi$ (Doc. 133)	<i>local</i>	6.67×10^{-5}
$K_{\text{frak}} = 1 - 100\xi$ (Doc. 133, A040)	<i>cumulative</i>	1.33×10^{-2}

Doc. 133 explicitly derives K_{frak} *not* from the local dimension but from the “cumulative effect” across the RG run from the Planck to the electroweak scale (17 orders of magnitude); the factor 100 is RG amplification (Doc. 190). The local dimension alone yields, per Doc. 133, “a vanishingly small correction – practically one”, computed $(D/3)^{D/2} = 0.99993$. The ratio of the two effects is exactly 200.

On notation, because this is easily slipped: the fractal dimension of FFGFT is $D_f^{\text{space}} = 3 - \xi$ – the *spatial* one, not a four-dimensional quantity. The exponent $3/2$ in $K_{\text{frak}}^{3/2}$ (Doc. 018) is likewise justified as half the *spatial* dimension. It is three dimensions, not four; a quantity $D_f = 4 - \xi$ is unknown to the corpus.

Hence the assignment – and it coincides with the boundary of Ch. D2 (rolled up versus unrolled): a Laplacian on the compact T^4 is *local*. It cannot carry a cumulative quantity; K_{frak} does not enter through the operator but through the **bridge constants** (E_0, ν) in scale conversion – exactly as R72 shows for the two E_0 values (ratio $1/\sqrt{K_{\text{frak}}}$). The consequence falls into three very unequal parts.

J.1 Untouched: everything anchor-free

By the assignment rule (register R70, Doc. 313), K_{frak} applies to absolute quantities and cancels for ratios of the same level. Untouched therefore:

- all **counts** – 24, 8, $N(n)$, 8×3 , 4×6 : integer combinatorics, no length;
- the **congruences** $N(n) \equiv 0 \pmod{3}$ and $\pmod{6}$: group theory, no metric content;
- the **character split** $N/3$ per sector;
- all **ratios** E/E_{min} and the crossing radii $\sqrt{2}$, $\sqrt{3}$ – structure over structure, the factor cancels.

This is the real reason why Ch. H is called “anchor-free results”: those statements are fractal-invariant.

J.2 One factor: all absolute energies

With an anchor, $E = \hbar H_0 |k|$ (ground mode = $\hbar H_0$, see Ch. B together with Doc. 312). These values are bare and carry the factor:

	bare	with K_{frak}
first D4 shell ($ k = \sqrt{2}$)	$3.230 \times 10^{-52} \text{ J}$	$3.187 \times 10^{-52} \text{ J}$
pure winding ($ k = 2$)	$4.567 \times 10^{-52} \text{ J}$	$4.507 \times 10^{-52} \text{ J}$

Throughout -1.33% . The same holds for the Casimir energies of Ch. H and for the spectral gap itself. Bookkeeping, settled; not structurally interesting.

Important for the assignment: the factor sits *not* in the spectrum but in the *anchor*. The dimensionless eigenvalues $|k|^2$ remain unchanged; what is corrected is the conversion to SI via $\hbar H_0$ – a scale bridge, that is, precisely the route along which a cumulative quantity may act. Anyone looking only at ratios sees none of it (A040 rule, R72).

J.3 The operator, not the conversion: the perturbation calculation [K]

The third part cannot be settled by multiplication. At $D_f^{\text{space}} = 3 - \xi$ the operator is *no longer an exactly smooth Laplacian* – and exact symmetry is precisely what carries the degeneracies. Only the *local* effect is relevant here: K_{frak} is cumulative per Doc. 133 and therefore does not belong on a local operator, so $O(100\xi)$ is ruled out as a perturbation strength.

Perturbation strength	$\Delta E/E$	ΔE at $E = \hbar H_0$
local, $(D/3)^{D/2}$ (Doc. 133)	6.7×10^{-5}	$1.5 \times 10^{-56} \text{ J}$
$O(\xi)$ (upper bound)	1.3×10^{-4}	$3.0 \times 10^{-56} \text{ J}$

The perturbation thus lies **two orders of magnitude below** what an estimate via K_{frak} would give.

The 24-fold degeneracy would then no longer be a degeneracy but a multiplet with fine structure. Which sub-multiplets it decomposes into is decided by the *symmetry of the perturbation* – and, as the calculation shows, via *representation theory*, not via the orbit structure:

Perturbation respects	Multiplet structure of the 24 (computed)
full $\text{Aut}(D_4)$ (c_g norm-dependent only)	$9 + 8 + 4 + 2 + 1$
triality (Z_3) and -1	$8 \times 1 + 8 \times 2$
only Z_3	$8 \times 1 + 8 \times 2$
only -1	24×1
neither	24×1

Computed (script 4): first-order perturbation theory on the 24×24 matrix $\langle k|V|k' \rangle = c_{k-k'}$, symmetry classes via orbit averaging of the coefficients, three independent random perturbations per class – the multiplet structure is seed-independent and identical in all cases.

The obvious expectation is wrong. One assumes the orbits stay together (Z_3 and -1 : the 24 as a whole; only Z_3 : 8 triple orbits; only -1 : 4 six-blocks). In fact multiplet sizes follow the *irreducible representations* of the respected symmetry group, not the orbits. Concretely: (i) every position-dependent perturbation breaks the translation invariance of the torus – and *that* carried the 24-fold degeneracy, not the point symmetry. (ii) Z_3 is abelian, all irreps one-dimensional; the doublets are the antiunitary pairing of the conjugate characters $\chi_1 \leftrightarrow \chi_2$, while the χ_0 sector decomposes into 8 singlets. (iii) -1 alone enforces nothing (free Z_2 action, one-dimensional characters). (iv) Even the maximally symmetric perturbation does not keep the 24 together: $24 = 9 \oplus 8 \oplus 4 \oplus 2 \oplus 1$, consistent with the irrep dimensions of $W(F_4)$ on the root shell [K numerically; the irrep assignment is B, without a character computation].

This closes the open point – and defuses it: qualitatively it still holds that only as much of the 24-fold degeneracy survives as the perturbation symmetry provides in irrep dimension (at best $9 + 8 + 4 + 2 + 1$, generically nothing). Quantitatively, however, the relevant perturbation is $\sim 10^{-5}$ relative, not 10^{-2} : the splitting would be fine structure far below any readability, and the container structure (4×6 , Ch. G) remains practically intact.

Chapter balance. *Bare* are only the SI absolute values, and via the anchor, not via the operator. The rolled-up quantities of this document – degeneracies, ratios, congruences, containers, Casimir and heat-kernel ratios – are **not** bare but correct: K_{frac} does not act there structurally. This is the same boundary as in Ch. D2, read from the other side.

Chapter K

Status overview

Statement	Content	Status
Translation	geometry $\rightarrow$ spectrum + fibre; lattice becomes mode index, degeneracies = theta series	[K]
Distinguishability	Z^4 from norm 1 (8), D4 from norm 2 (24); odd norms empty for D4	[K]
12 + 12	first shell exactly FCC + winding; higher shells asymmetric (6/18, 8/88)	[K]
Anisotropy	$E_{\text{wind}} = 1 + 1/r^2$; crossings only at $r = \sqrt{2}$ and $\sqrt{3}$; three regimes	[K]
Triality	$ \text{Aut}(D4) = 1152$, quotient $6 = S_3 $; 80 order-3 elements in three classes	[K]
Orbifold	16 elements with $ \det(1 - A) = 9$: T^4/Z_3 is a lattice symmetry, not an extra assumption	[K]
Circulant	Z_3 acts on every triple orbit with $\{1, \omega, \omega^2\} = \mathbb{C}^3$ fibre structure	[K]
5 $\nmid$ 1152	no order-5 automorphisms; 2/9 is not a lattice invariant (denominators $\{1, 2, 3, 6\}$)	[K] theorem
Doubling	interface $A_5 \cap \text{Aut}(D4) = C_3$, occupied; $24 = 4 \times 6$ as containers	[K]
Labelling	3 versus 3' is lattice-blind (difference only on 5-classes)	[K] boundary
Congruences	$N(n) \equiv 0 \pmod{3}$ and $\pmod{6}$ on all 60 shells checked, with proof ground	[K] theorem
Selection	containers and equidistribution are generic – the structure cannot distinguish any shell	[K]
Casimir	$E(Z^4) = -0.9321$ against $E(D4) = -0.8686$; ordering reversed relative to $s > 0$	[K], field-content dependent
Heat kernel	Weyl term identical, difference 4×10^{-27} at $t = 0.05$: leading order is lattice-blind	[K]
Spectral theorem	"spectrum determines lattice" false in 4D (Schiemann); uncritical for D4, but not a rule	[K] boundary
Reciprocity	$\lambda_4 \cdot m = 2\pi$ exact as a place relation (de Broglie); temporally $2\pi/\gamma \leq 2\pi$; for mixtures $\geq 2\pi$ (Jensen) – exact only at rest and sharp	[K]

Statement	Content	Status
Closure case	rolled up, no fractal correction (k_4 topological); fork A/B/C spectrally invisible, Chs. E–H case-independent	[K]
Perturbation multiplets	splitting follows irrep dimensions, not orbits: $9+8+4+2+1$ (full), $8 \times 1+8 \times 2$ (Z_3), 24×1 (generic)	[K]
Fractal status	$D_f^{\text{space}} = 3 - \xi$ local (6.7×10^{-5}) versus K_{frak} cumulative (1.33 %, RG run, Doc. 133): bare are only SI absolute values via the anchor, not the rolled-up spectrum	[K]

Balance: in Hilbert space the lattice becomes flat but not inconsequential. The kissing number becomes the degeneracy, triality becomes the orbifold and the fibre, the radius ratio becomes the level splitting. The *division of labour* is determined just as sharply: five-fold symmetry – and with it φ and $2/9$ – cannot be carried by the lattice *itself*, and that by theorem ($5 \nmid 1152$), not for want of computational effort. This is a statement about the *layer*, not about the quantity: on the fibre, $2/9$ is explicitly admissible, since the interface $A_5 \cap \text{Aut}(D_4) = C_3$ is occupied by triality (Ch. G), and with $24 = 4 \times 6$ the lattice even provides the containers for the doubling $3 \oplus 3'$. What is excluded is therefore not $2/9$ but only its derivation *from* lattice invariants – the phase demonstrably enters one layer higher. And the container structure is generic: it provides, it does not select.

Chapter L

Check scripts

Three scripts, all deterministic, all controls passed:

- `d4_skript_1_spektrum_deformation.py` – theta series $D4/Z^4$, k_4 split, deformation r , exact crossing radii (standard library only).
- `d4_skript_2_trialitaet_orbifold_phasen.py` – $\text{Aut}(D4) = 1152$, order-3 classification (computed exactly with $2A$), $|\det(1 - A)| = 9$, circulant, phase test $2/9$, order distribution, $24 = 4 \times 6$ (numpy).
- `d4_skript_4_stoerung_reziprozitaet.py` – first-order perturbation theory on the 24 shell (multiplet structure per symmetry class, 3 seeds each) and reciprocity $T \cdot m$ (mode-wise; Jensen for mixtures) (numpy, mpmath).
- `d4_skript_3_schalen_casimir.py` – shell theorems, Epstein zeta at $s = -1/2$, heat kernel (numpy, mpmath).

Controls against known values: θ_{D4} , $|W(F_4)| = 1152$, $2^4 \cdot 4! = 384$, $|W(D4)| = 192$, orbifold fixed-point count $3^2 = 9$, Jacobi identity for Z_7^4 .

Registration: entry into Doc. 190 will be done separately, should the delimitation against Doc. 293 be booked there.